

GitHub repository where all documentation & code can be found. [billbill100 \(github.com\)](https://github.com/billbill100)

ESP32 3d print case

V1.2 10/03/2026

To print the case.

There are four parts to print.

- 1) The main box (top sides & front).
- 2) Case bottom.
- 3) The rear panel, of which there are three options to choose from.
 - a) plain, which is the easiest to print.
 - b) the version with indented legends, labelling the sockets.
 - c) Third version has flat lettering. This requires a little more skill to print as the lettering needs to be selected and given a different colour to the box. Then the slicer needs setting so it prints all of the text first, followed by the rest of the rear panel.
- 4) The final part is the expansion board frame if using the screw-terminal expansion board from AliExpress, or pcb spacer, if using the pcb expansion board.

There is also a blanking cap to print, for unused holes.

To Assemble. Four M3 X 25mm bolts go through the four holes in the front of the case.
On the inside, the TFT screen fits onto the bolts.

The expansion board frame (legs down) fit onto these bolts and it is all secured together with four washers and nuts.

The screw-terminal expansion board is mounted to the frame by drilling four small holes in the expansion frame, to match that of the expansion board itself, & attached with 2mm screws. The holes are not pre-drilled as expansion board hole positions can vary.

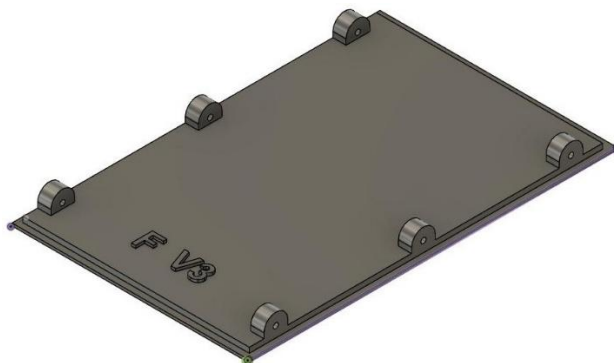
If using the pcb expansion board, the 3d printed pcb spacer is bolted to the lower holes on the pcb. M3 x 12MM bolts ideal, using the round holes. The elongated holes on the 3d printed spacer fit over the lower tft mounting bolts.

The 3d case is designed to be screwed together using 2mm self-tapping screws.

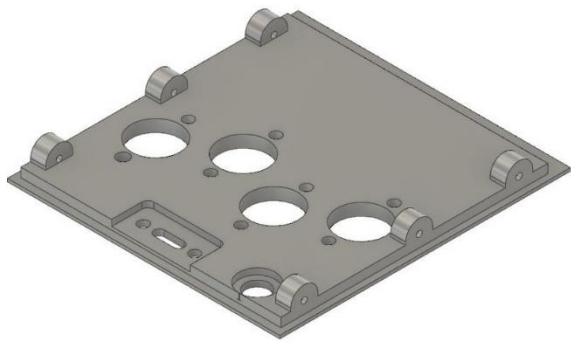
There is a lot of wiring to be connected (the pcb version negates some of this) and so it can be easier not to mount the expansion board until wiring is complete.



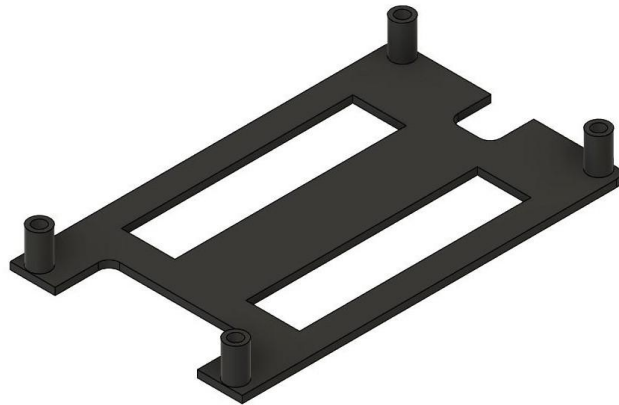
Main case



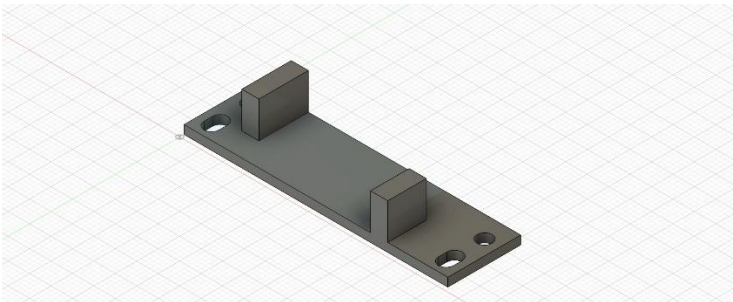
Rear



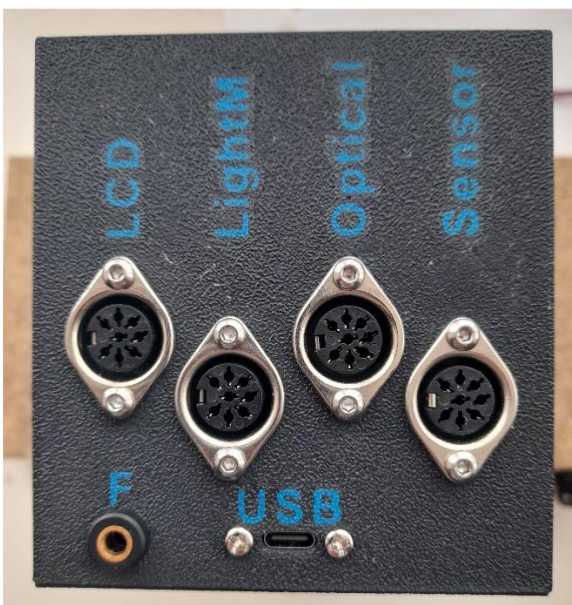
Back



Expansion Frame (if using screw-terminal expansion board)



Pcb spacer (if using pcb expansion board)



rear, showing option 3, with the colour printed legends.

